EXTERNAL HALL / ABI: 3.3V or 5V logic -> 3.3V Schmitt buffer -> reset isolation

Joint feedback supply: fixed, protected 5V shared with J502 fan.
J601.1 = 5V output; J601.2/6 = GND. No power input.
For a 5V Hall/ABI encoder; no voltage selector.
Sensor + pull-ups target <=25mA; shared accessory target <=300mA.

D601 rail isolated from supplies.
C604 is required; no 3V3 clamp tie.

INPUT OPTIONS
Default: 3.3V / 5V push-pull; pull-ups DNP.
Open-drain: fit R608-R610 (4.7k to protected 5V).
Do not fit pull-ups for externally driven push-pull.
100k pull-downs define disconnected inputs.
Single-ended only; common ground required.

D601
TPD3E001DRYR
NC
GND
VCC
IO3
IO2
IO1
FB_ESD_RAIL
C604
100nF 50V

J601
MP
MOUNT
EXT ENCODER / HALL

R608
4.7k DNP
R609
4.7k DNP
R610
4.7k DNP
R605
100R
R606
100R
R607
100R
R615
100k
R616
100k
R617
100k

U603A
U603C
U603B
74LVC3G17

U602A
TMUX1574PW
TP1016
FEEDBACK_B_H2
TP1015
FEEDBACK_A_H1
FEEDBACK_I_H3

U604
TPS2553DBVR
IN
EN
ILIM
OUT
FAULT
GND
5V_SYS
EXT_VCC
EXT_POWER_FAULT_N
C606
4.7uF
R618
49.9k 1%
R604
10k
C607
100nF 16V X7R

U604: R618=49.9k 1% =>520mA nominal shared limit.
EXT_VCC feeds J601 sensor + J502 fan + U503 PWM buffer.
EXT_POWER_FAULT_N -> PC13; shared branch fault.
Stop torque on fault or external-feedback loss; verify fan startup margin.

U602B
TMUX1574PW
VBUS_MUX_IN
VBUS_SENSE

U603D
74LVC3G17
C605
100nF

INPUT_MUX_ENABLE_N
EXT_FEEDBACK_SELECT
R305
10k
R306
10k
U602C
TMUX1574PW
SEL
EN_N
VDD
C603
100nF
R307
10k

Reset: mux disabled, PB8/BOOT0 held low by R307.
EN_N=0 enables VM sensing; wait >=2ms before ADC read.
SEL=0 grounds TIM4 inputs; SEL=1 connects external Hall/ABI.
Onboard absolute-angle SPI is independent of this switch.

3.3V / 5V external inputs; interface revision
ShiroFOC
Sheet: /
File: 09_Feedback_Selection.kicad_sch
Title: External Hall / ABI conditioning and reset isolation
Size: A3 | Date: 2026-09-26 | **Rev: A1-Drawing**
KiCad E.D.A. 9.0.4 | Id: 1/1